Electromagnetism & Electrodynamics HW3

Daniel Haim Breger, 316136944

Technion

(Date: June 6, 2026)

1 Moving Point Charge

A point charge q is moving at a constant velocity $\mathbf{v} = v\hat{\mathbf{z}}$ and is at the origin at $t = 0$, in the lab frame S. In the charge's rest frame, S', the electric and magnetic fields are given by

$$\mathbf{E}' = \frac{q\hat{\mathbf{r}'}}{r'^2}, \quad \mathbf{B}' = 0 \tag{1}$$

using the field transformation rule, find the electric and magnetic field in the lab frame S

In cartesian coordinates the fields are

$$E' = q\frac{\hat{x}' + \hat{y}' + \hat{z}'}{x'^2 + y'^2 + z'^2} = q\frac{x' + y' + z'}{(x'^2 + y'^2 + z'^2)^{3/2}} \qquad B' = 0 \tag{2}$$

giving us the fields tensor

$$F'_{\mu\nu} = \frac{q}{(x'^2 + y'^2 + z'^2)^{3/2}} \begin{pmatrix} & x' & y' & z' \\ -x' & & & \\ -y' & & & \\ -z' & & & \end{pmatrix} \tag{3}$$

and using the lorentz boost matrix

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & & & -\gamma\beta \\ & 1 & & \\ & & 1 & \\ -\gamma\beta & & & \gamma \end{pmatrix} \qquad (\Lambda^{-1})^\mu_\nu = \begin{pmatrix} \gamma & & & \gamma\beta \\ & 1 & & \\ & & 1 & \\ \gamma\beta & & & \gamma \end{pmatrix} \tag{4}$$

we get

$$F_{\mu\nu} = \Lambda^\sigma_\mu F'_{\sigma\varepsilon}\Lambda^\varepsilon_\nu \tag{5}$$

$$= \left[\Lambda^T F \Lambda\right]_{\mu\nu} \tag{6}$$

$$= \frac{q}{(x'^2+y'^2+z'^2)^{3/2}} \begin{pmatrix} \gamma & & & -\gamma\beta \\ & 1 & & \\ & & 1 & \\ -\gamma\beta & & & \gamma \end{pmatrix} \begin{pmatrix} & x' & y' & z' \\ -x' & & & \\ -y' & & & \\ -z' & & & \end{pmatrix} \begin{pmatrix} \gamma & & & -\gamma\beta \\ & 1 & & \\ & & 1 & \\ -\gamma\beta & & & \gamma \end{pmatrix} \tag{7}$$

$$= \frac{q}{(x'^2+y'^2+z'^2)^{3/2}} \begin{pmatrix} \gamma & & & -\gamma\beta \\ & 1 & & \\ & & 1 & \\ -\gamma\beta & & & \gamma \end{pmatrix} \begin{pmatrix} -\gamma\beta z' & x' & y' & \gamma z' \\ -\gamma x' & & & \gamma\beta x' \\ -\gamma y' & & & \gamma\beta y' \\ -\gamma z' & & & \gamma\beta z' \end{pmatrix} \tag{8}$$

$$= \frac{q}{(x'^2+y'^2+z'^2)^{3/2}} \begin{pmatrix} -\gamma^2\beta z' + \gamma^2\beta z' & \gamma x' & \gamma y' & \gamma^2 z' - \gamma^2\beta^2 z' \\ -\gamma x' & & & \gamma\beta x' \\ -\gamma y' & & & \gamma\beta y' \\ \gamma^2\beta^2 z' - \gamma^2 z' & -\gamma\beta x' & -\gamma\beta y' & -\gamma^2\beta z' + \gamma^2\beta z' \end{pmatrix} \tag{9}$$

$$= \frac{q}{(x'^2+y'^2+z'^2)^{3/2}} \begin{pmatrix} & \gamma x' & \gamma y' & z' \\ -\gamma x' & & & \gamma\beta x' \\ -\gamma y' & & & \gamma\beta y' \\ z' & -\gamma\beta x' & -\gamma\beta y' & \end{pmatrix} \tag{10}$$

converting the coordinates themselves

$$x'^\mu = \Lambda^\mu{}_\nu x'^\nu = \begin{pmatrix} \gamma & & & -\gamma\beta \\ & 1 & & \\ & & 1 & \\ -\gamma\beta & & & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \gamma ct - \gamma\beta z \\ x \\ y \\ \gamma z - \gamma\beta ct \end{pmatrix} = \begin{pmatrix} \gamma(ct - \beta z) \\ x \\ y \\ \gamma(z - vt) \end{pmatrix} \tag{11}$$

so the fields tensor in the lab frame is

$$F_{\mu\nu} = \frac{q}{(x^2+y^2+(\gamma(z-vt))^2)^{3/2}} \begin{pmatrix} & \gamma x & \gamma y & \gamma(z-vt) \\ -\gamma x & & & \gamma\beta x \\ -\gamma y & & & \gamma\beta y \\ \gamma(z-vt) & -\gamma\beta x & -\gamma\beta y & \end{pmatrix} \tag{12}$$

giving us the fields

$$E = \frac{\gamma q}{(x^2+y^2+(\gamma(z-vt))^2)^{3/2}} \begin{pmatrix} x \\ y \\ z - vt \end{pmatrix} \qquad B = \frac{\gamma\beta q}{(x^2+y^2+(\gamma(z-vt))^2)^{3/2}} \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} \tag{13}$$

2 Fields and a Particle

The lab frame has uniform electric and magnetic fields

$$\mathbf{E} = E_0\hat{\mathbf{z}} \quad \mathbf{B} = B_0\hat{\mathbf{x}} \tag{14}$$

1. Use the formula for the lorentz transformation for EM fields and the find the velocity of the coordinate system in which the electric field becomes zero. What is the magnetic field in this frame? What is the condition on E_0, B_0 to move to this frame?
2. A point charge q with mass m is released at rest from the origin in the lab frame. Write the equations of motion in the frame with no electric field and find the momentum 4-vector
3. Find the trajectory as a function of the self-time and of the lab frame time
4. What is the time in which the particle performs a revolution in the yz plane as measured in the frame with no electric field.
5. Use the results and write the trajectory of the particle in the lab frame depending on the self time

2.1 Zero electric field

The fields tensor is

$$\begin{pmatrix} 0 & 0 & 0 & E_0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -B_0 \\ -E_0 & 0 & B_0 & 0 \end{pmatrix} \tag{15}$$

and the lorentz boost is

$$\begin{pmatrix} \gamma & 0 & \beta\gamma & 0 \\ 0 & 1 & 0 & 0 \\ \beta\gamma & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{16}$$

so the transformed tensor is

$$\Lambda F \Lambda^T = \begin{pmatrix} 0 & 0 & 0 & \gamma E_0 - \beta\gamma B_0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \beta\gamma E_0 - \gamma B_0 \\ -\gamma E_0 + \beta\gamma B_0 & 0 & -\beta\gamma E_0 + \gamma B_0 & 0 \end{pmatrix} \tag{17}$$

the condition we have therefore is

$$\gamma E_0 - \beta\gamma B_0 = 0 \implies \beta = \frac{E_0}{B_0} \implies v = \frac{E_0}{B_0} c\hat{\mathbf{y}} \tag{18}$$

The magnetic field is then

$$\mathbf{B} = \gamma\left(B_0 - \frac{E_0^2}{B_0}\right)\hat{\mathbf{x}} = \gamma\left(1-\beta^2\right)B_0\hat{\mathbf{x}} = \frac{B_0}{\gamma}\hat{\mathbf{x}} \tag{19}$$

we can see that the requirement to be able to move to this frame is that the magnetic field is not zero, and that $B_0 > E_0$.

2.2 Charge released from rest

The equation of motion for a charged particle is

$$m\frac{\mathrm{d}^2 x_\mu}{\mathrm{d}\tau^2} - \frac{q}{c}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}F_{\mu\nu} = 0 \tag{20}$$

which can be written as

$$\frac{\mathrm{d}p_\mu}{\mathrm{d}\tau} - \frac{q}{c}u^\nu F_{\mu\nu} = 0 \tag{21}$$

multiplying and dividing by the mass

$$\frac{\mathrm{d}p_\mu}{\mathrm{d}\tau} - \frac{q}{mc}mu^\nu F_{\mu\nu} = \frac{\mathrm{d}p_\mu}{\mathrm{d}\tau} - \frac{q}{mc}p^\nu F_{\mu\nu} = 0 \tag{22}$$

raising the index

$$\eta_{\mu\sigma}\frac{\mathrm{d}p^\sigma}{\mathrm{d}\tau} = \frac{q}{mc}p^\nu F_{\mu\nu} \tag{23}$$

multiplying by the dual metric

$$\eta^{\mu\rho}\eta_{\rho\sigma}\frac{\mathrm{d}p^\sigma}{\mathrm{d}\tau} = \frac{q}{mc}\eta^{\mu\rho}p^\nu F_{\mu\nu} \tag{24}$$

simplifying

$$\delta^\mu{}_\sigma\frac{\mathrm{d}p^\sigma}{\mathrm{d}\tau} = \frac{q}{mc}\eta^{\mu\rho}p^\nu F_{\mu\nu} \tag{25}$$

$$\frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = \frac{q}{mc}\eta^{\mu\rho}p^\nu F_{\mu\nu} \tag{26}$$

in matrix form

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} = \frac{q}{mc}\eta^{-1}Fp \tag{27}$$

inserting the fields tensor

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} = \frac{q}{mc}\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}\begin{pmatrix} & & & \\ & & & \\ & & & -\frac{B_0}{\gamma} \\ & & \frac{B_0}{\gamma} & \end{pmatrix}p \tag{28}$$

which simplifies to

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} = \frac{q}{mc}\frac{B_0}{\gamma}\begin{pmatrix} & & & \\ & & & \\ & & & 1 \\ & & -1 & \end{pmatrix}p \tag{29}$$

we find the eigenvalues

$$\lambda_0 = 0, v_{0,1} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \qquad \lambda_{1,2} = \pm i, v_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ i \end{pmatrix}, v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -i \end{pmatrix} \tag{30}$$

which means the solutions are

$$p = c_0 v_0 + c_1 v_1 + c_2 e^{i\frac{qB_0}{mc}\tau} v_2 + c_3 e^{-i\frac{qB_0}{mc}\tau} v_3 = \begin{pmatrix} c_0 \\ c_1 \\ c_2 e^{i\frac{qB_0}{mc}\tau} + c_3 e^{-i\frac{qB_0}{mc}\tau} \\ \left(c_2 e^{i\frac{qB_0}{mc}\tau} - c_3 e^{-i\frac{qB_0}{mc}\tau}\right) i \end{pmatrix} \tag{31}$$

but the particle started at rest

$$p_0^\mu = \Lambda^\mu{}_\nu p_0^\nu = \begin{pmatrix} \gamma & & -\gamma\beta & \\ & 1 & & \\ -\gamma\beta & & \gamma & \\ & & & 1 \end{pmatrix} \begin{pmatrix} mc \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma mc \\ 0 \\ -\gamma\beta mc \\ 0 \end{pmatrix} \tag{32}$$

therefore

$$c_0 = \gamma mc \qquad c_1 = 0 \qquad c_2 + c_3 = -\gamma\beta mc \qquad c_2 - c_3 = 0 \tag{33}$$

giving us

$$c_2 = c_3 = -\frac{\gamma\beta mc}{2} = -\frac{\gamma mv}{2} \tag{34}$$

so the 4-momentum is

$$p^\mu(\tau) = \begin{pmatrix} \gamma mc \\ 0 \\ -\frac{\gamma mv}{2}\left(e^{i\frac{qB_0}{mc}\tau} + e^{-i\frac{qB_0}{mc}\tau}\right) \\ -\frac{i\gamma mv}{2}\left(e^{i\frac{qB_0}{mc}\tau} - e^{-i\frac{qB_0}{mc}\tau}\right) \end{pmatrix} = \gamma m \begin{pmatrix} c \\ 0 \\ -v\cos\frac{qB_0}{mc}\tau \\ v\sin\frac{qB_0}{mc}\tau \end{pmatrix} \tag{35}$$

2.3 Self time position

By definition

$$p'^\mu = m\frac{\mathrm{d}x'^\mu}{\mathrm{d}\tau} \tag{36}$$

so if we combine with a result from the previous section we get

$$m\frac{\mathrm{d}x'^\mu}{\mathrm{d}\tau} = \begin{pmatrix} \gamma mc \\ 0 \\ -\gamma mv\cos\frac{qB_0}{mc}\tau \\ \gamma mv\sin\frac{qB_0}{mc}\tau \end{pmatrix} \tag{37}$$

canceling terms and integrating over τ:

$$\int \frac{\mathrm{d}x'^{\mu}}{\mathrm{d}\tau}\, d\tau = \int \begin{pmatrix} \gamma c \\ 0 \\ -\gamma v \cos \frac{qB_0}{mc}\tau \\ \gamma v \sin \frac{qB_0}{mc}\tau \end{pmatrix} d\tau \tag{38}$$

therefore

$$x'^{\mu} = \begin{pmatrix} \gamma c\tau + c_0 \\ c_1 \\ -\frac{\gamma mvc}{qB_0} \sin \frac{qB_0}{mc}\tau + c_2 \\ -\frac{\gamma mvc}{qB_0} \cos \frac{qB_0}{mc}\tau + c_3 \end{pmatrix} \tag{39}$$

But since the particle started in the origin (in the lab's reference frame but this lorentz transforms to the origin as well in the particle's frame)

$$c_0 = 0 \qquad c_1 = 0 \qquad c_2 = 0 \qquad c_3 = \frac{\gamma mvc}{qB_0} \tag{40}$$

Thus the particle's path is

$$x'^{\mu} = \begin{pmatrix} \gamma ct \\ 0 \\ -\frac{\gamma mvc}{qB_0} \sin \frac{qB_0}{mc}\tau \\ -\frac{\gamma mvc}{qB_0} \cos \frac{qB_0}{mc}\tau + \frac{\gamma mvc}{qB_0} \end{pmatrix} = \begin{pmatrix} \gamma ct \\ 0 \\ -\frac{\gamma mvc}{qB_0} \sin \frac{qB_0}{mc}\tau \\ \frac{\gamma mvc}{qB_0}(1 - \cos \frac{qB_0}{mc}\tau) \end{pmatrix} \tag{41}$$

2.4 revolution time

By definition in the rest frame

$$T = \frac{2\pi}{\omega} = \frac{2\pi mc}{qB_0} \tag{42}$$

therefore in the lab frame

$$T' = \gamma T = \frac{2\gamma\pi mc}{qB_0} \tag{43}$$

2.5 Path in self time

We'll boost back

$$\begin{aligned} x^{\mu}(\tau) &= (\Lambda^{-1})^{\mu}{}_{\nu} x'^{\nu}(\tau) &(44)\\ &= \begin{pmatrix} \gamma & & \gamma\beta & \\ & 1 & & \\ \gamma\beta & & \gamma & \\ & & & 1 \end{pmatrix} \begin{pmatrix} \gamma ct \\ 0 \\ -\frac{\gamma mvc}{qB_0} \sin \frac{qB_0}{mc}\tau \\ \frac{\gamma mvc}{qB_0}(1 - \cos \frac{qB_0}{mc}\tau) \end{pmatrix} &(45)\\ &= \begin{pmatrix} \gamma^2 ct - \frac{\gamma^2 \beta mvc}{qB_0} \sin \frac{qB_0}{mc}\tau \\ 0 \\ \gamma^2 \beta ct - \frac{\gamma^2 mvc}{qB_0} \sin \frac{qB_0}{mc}\tau \\ \frac{\gamma mvc}{qB_0}(1 - \cos \frac{qB_0}{mc}\tau) \end{pmatrix} &(46) \end{aligned}$$

3 Capacitor

A capacitor is made of two square sides with side length a at a distance of d apart such that $d \ll a$. The top plate is charged with $-Q$ and the bottom with $+Q$. We'll define the axes such that its center coincides with the center of the capacitor, etc. Find the electric and magnetic fields between the capacitor's sides in

1. a reference frame moving at $\mathbf{v} = u\hat{\mathbf{x}}$
2. a reference frame moving at $\mathbf{v} = u\hat{\mathbf{z}}$

I am working in Gaussian units, in which $\varepsilon_0 = \frac{1}{4\pi}$.
The fields inside a capacitor in its own rest frame are

$$F_{\mu\nu} = 4\pi\sigma \begin{pmatrix} & & & 1 \\ & & & \\ & & & \\ -1 & & & \end{pmatrix} \tag{47}$$

where

$$\sigma = \frac{Q}{a^2} \tag{48}$$

3.1 x direction

We'll perform a boost

$$\begin{aligned} F'_{\mu\nu} &= (\Lambda^{-1})_\mu{}^\sigma F_{\sigma\varepsilon} \Lambda^\varepsilon{}_\nu && (49) \\ &= \left[(\Lambda^{-1})^T F (\Lambda^{-1}) \right]_{\mu\nu} && (50) \\ &= 4\pi\sigma \begin{pmatrix} \gamma & \gamma\beta & & \\ \gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} & & & 1 \\ & & & \\ & & & \\ -1 & & & \end{pmatrix} \begin{pmatrix} \gamma & \gamma\beta & & \\ \gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix} && (51) \\ &= 4\pi\sigma \begin{pmatrix} \gamma & \gamma\beta & & \\ \gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} & & & 1 \\ & & & \\ & & & \\ -\gamma & -\gamma\beta & & \end{pmatrix} && (52) \\ &= 4\pi\gamma\sigma \begin{pmatrix} & & & 1 \\ & & & \beta \\ & & & \\ -1 & -\beta & & \end{pmatrix} && (53) \end{aligned}$$

giving us the fields

$$E' = 4\pi\gamma\sigma\hat{\mathbf{z}} \qquad\qquad B' = 4\pi\gamma\beta\sigma\hat{\mathbf{y}} \tag{54}$$

inside the capacitor.

3.2 z direction

same as before

$$
\begin{aligned}
F'_{\mu\nu} &= (\Lambda^{-1})_\mu{}^\sigma F_{\sigma\varepsilon} \Lambda^\varepsilon{}_\nu && (55) \\
&= \left[(\Lambda^{-1})^T F (\Lambda^{-1}) \right]_{\mu\nu} && (56) \\
&= 4\pi\sigma \begin{pmatrix} \gamma & & & \gamma\beta \\ & 1 & & \\ & & 1 & \\ \gamma\beta & & & \gamma \end{pmatrix} \begin{pmatrix} & & & 1 \\ & & & \\ & & & \\ -1 & & & \end{pmatrix} \begin{pmatrix} \gamma & & & \gamma\beta \\ & 1 & & \\ & & 1 & \\ \gamma\beta & & & \gamma \end{pmatrix} && (57) \\
&= 4\pi\sigma \begin{pmatrix} \gamma & & & \gamma\beta \\ & 1 & & \\ & & 1 & \\ \gamma\beta & & & \gamma \end{pmatrix} \begin{pmatrix} \gamma\beta & & & \gamma \\ & & & \\ & & & \\ -\gamma & & & -\gamma\beta \end{pmatrix} && (58) \\
&= 4\pi\gamma\sigma \begin{pmatrix} \gamma^2\beta - \gamma^2\beta & & & \gamma^2 - \gamma^2\beta^2 \\ & & & \\ & & & \\ \gamma^2\beta^2 - \gamma^2 & & & \gamma^2\beta - \gamma^2\beta \end{pmatrix} && (59) \\
&= 4\pi\gamma\sigma \begin{pmatrix} & & & \gamma^2 - \gamma^2\beta^2 \\ & & & \\ & & & \\ \gamma^2\beta^2 - \gamma^2 & & & \end{pmatrix} && (60) \\
&= 4\pi\gamma\sigma \begin{pmatrix} & & & 1 \\ & & & \\ & & & \\ -1 & & & \end{pmatrix} && (61) \\
& && (62)
\end{aligned}
$$

Therefore the fields are

$$
E' = 4\pi\gamma\sigma\hat{\mathbf{z}} \qquad\qquad B' = 0 \tag{63}
$$

and this field again only exists inside the capacitor, but because of the relativistic length contraction it is reduced to

$$
-\frac{d}{2\gamma} \leq z' + ut' \leq \frac{d}{2\gamma} \tag{64}
$$